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Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (R, S) / S1 (PT) (S, FE) Examination December 2023 (2019 Scheme)



Course Code: CST205

Course Name: OBJECT ORIENTED PROGRAMMING USING JAVA

Duration: 3 Hours

Max. Marks: 100

PART A

Answer all questions. Each question carries 3 marks

Marks

- | | | |
|----|---|-----|
| 1 | Describe the role of JIT compiler. | (3) |
| 2 | Why Java is said to be robust? | (3) |
| 3 | List the properties of a constructor. | (3) |
| 4 | Discuss the use of public, private and protected access specifiers. | (3) |
| 5 | Distinguish between checked and unchecked exceptions. | (3) |
| 6 | Explain any three character stream classes in Java | (3) |
| 7 | Explain the different methods of creating threads in Java. | (3) |
| 8 | List any three event listeners and the corresponding source object. | (3) |
| 9 | Describe the features of Swing. | (3) |
| 10 | Explain about the containers in Swing. | (3) |

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

- | | | | |
|----|---|---|-----|
| 11 | a | Distinguish between function oriented and object oriented design paradigms with an example. | (8) |
| | b | Draw a UML activity diagram for an order processing system. | (6) |

OR

- | | | | |
|----|---|--|-----|
| 12 | a | Explain any four object oriented features of Java with suitable examples. | (8) |
| | B | Draw a UML class diagram which shows structure and attributes of an ATM with entities Bank, ATM, Debit Card, Customer, Account, ATM Transaction. | (6) |

Module 2

- | | | | |
|----|---|---|-----|
| 13 | a | Create a class Person with attributes name, age, address and a method display() to display the details. Create a subclass Student with attributes rollno, mark1, mark2 and mark3. Override display() method and calculate the | (8) |
|----|---|---|-----|

grade. Create another subclass Faculty with attributes faculty_id, department, basic_pay and DA. Override display() method to calculate total salary as (basic_pay+DA)+70% of (basic_pay+DA). Create instances of Student and Faculty and display the details.

- b Explain method overloading with an example. (6)

OR

- 14 a Explain dynamic method dispatch with an example (8)
b Discuss the uses of final keyword citing examples. (6)

Module 3

- 15 a Create a package reversepackage. Add a class Reverse in it with a method reverse() to print the reverse of a string without using built-in methods. Create a class outside the package and use this method to reverse a string. (8)
b Explain object stream serialization with sample code. (6)

OR

- 16 a How do you create user defined exceptions in Java? Write a Java program to find the average of n numbers entered as command line arguments. Raise a user defined exception if the average exceeds 100. (8)
b Explain the use of interface in Java. (6)

Module 4

- 17 a Write a multithreaded Java program with two threads. One thread generates even numbers from 1 to 100 and another thread generates odd numbers from 1 to 100. Ensure that the threads do not interfere. (8)
b Discuss the methods for String comparison with examples. (6)

OR

- 18 a Explain how event handling is done in Java. (8)
b How do you access the elements in an ArrayList? Illustrate with an example (6)

Module 5

- 19 a Explain with sample code the steps in establishing JDBC connectivity. (8)
b Describe the architecture of JDBC with a neat diagram. (6)

OR

- 20 a Write a GUI program to perform arithmetic operations of two numbers. (8)
b Discuss any three layout managers in Swing. (6)